

AI Code Review Report

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Overall Quality Score

56/100

Score Breakdown

Metric	Score
Readability	40/100
Security	50/100
Performance	92/100
Maintainability	50/100
Documentation	50/100

Complexity: Medium

Issues Found

Bugs

- Line 60: Potential infinite loop detected (*high*)
- Line 164: Division by zero detected (*critical*)

Security Issues

- Line 16: Hardcoded credential found (*high*)
- Line 52: Potential Command Injection vulnerability (*critical*)
- Line 79: eval/exec usage detected (*critical*)
- Line 84: Pickle is insecure for untrusted data (*high*)
- Line 89: Weak cryptographic algorithm "md5" used (*high*)

Performance Issues

- Line 112: Using range(len()) - consider using enumerate() (*low*)

Style Issues

- Line 31: Missing docstring (*medium*)
- Line 32: Missing docstring (*medium*)
- Line 43: Missing docstring (*medium*)
- Line 51: Missing docstring (*medium*)
- Line 55: Missing docstring (*medium*)
- Line 59: Missing docstring (*medium*)
- Line 66: Magic number 50 found (*low*)
- Line 70: Missing docstring (*medium*)
- Line 71: Magic number 42 found (*low*)
- Line 78: Missing docstring (*medium*)
- Line 82: Missing docstring (*medium*)
- Line 88: Missing docstring (*medium*)
- Line 93: Missing docstring (*medium*)
- Line 103: Missing docstring (*medium*)
- Line 110: Missing docstring (*medium*)
- Line 118: Line is 129 characters (max 100) (*low*)
- Line 118: Missing docstring (*medium*)
- Line 122: Magic number 50 found (*low*)
- Line 124: Magic number 75 found (*low*)
- Line 129: Missing docstring (*medium*)
- Line 135: Missing docstring (*medium*)
- Line 142: Missing docstring (*medium*)

- Line 146: Missing docstring (*medium*)
- Line 150: Line is 189 characters (max 100) (*low*)
- Line 153: Missing docstring (*medium*)
- Line 155: Magic number 20 found (*low*)
- Line 156: Magic number 30 found (*low*)
- Line 157: Magic number 40 found (*low*)
- Line 162: Missing docstring (*medium*)
- Line 185: Missing docstring (*medium*)
- Line 190: Missing docstring (*medium*)
- Line 196: Missing docstring (*medium*)
- Line 198: Magic number 20 found (*low*)
- Line 211: Missing docstring (*medium*)
- Line 237: Missing docstring (*medium*)
- Line 238: Missing docstring (*medium*)
- Line 241: Missing docstring (*medium*)
- Line 257: Missing docstring (*medium*)
- Line 266: Missing docstring (*medium*)
- Line 267: Missing docstring (*medium*)
- Line 278: Magic number 11 found (*low*)
- Line 279: Magic number 12 found (*low*)
- Line 280: Magic number 13 found (*low*)
- Line 281: Magic number 14 found (*low*)
- Line 282: Magic number 15 found (*low*)
- Line 283: Magic number 16 found (*low*)
- Line 284: Magic number 17 found (*low*)
- Line 285: Magic number 18 found (*low*)
- Line 286: Magic number 19 found (*low*)
- Line 287: Magic number 20 found (*low*)
- Line 288: Magic number 21 found (*low*)
- Line 289: Magic number 22 found (*low*)
- Line 290: Magic number 23 found (*low*)
- Line 291: Magic number 24 found (*low*)
- Line 292: Magic number 25 found (*low*)
- Line 293: Magic number 26 found (*low*)
- Line 295: Missing docstring (*medium*)
- Line 306: Magic number 11 found (*low*)
- Line 307: Magic number 12 found (*low*)
- Line 308: Magic number 13 found (*low*)
- Line 309: Magic number 14 found (*low*)
- Line 310: Magic number 15 found (*low*)

- Line 311: Magic number 16 found (*low*)
- Line 312: Magic number 17 found (*low*)
- Line 313: Magic number 18 found (*low*)
- Line 314: Magic number 19 found (*low*)
- Line 315: Magic number 20 found (*low*)
- Line 316: Magic number 21 found (*low*)
- Line 317: Magic number 22 found (*low*)
- Line 318: Magic number 23 found (*low*)
- Line 319: Magic number 24 found (*low*)
- Line 320: Magic number 25 found (*low*)
- Line 321: Magic number 26 found (*low*)
- Line 322: Magic number 27 found (*low*)
- Line 323: Magic number 28 found (*low*)
- Line 324: Magic number 29 found (*low*)
- Line 325: Magic number 30 found (*low*)
- Line 326: Magic number 31 found (*low*)
- Line 327: Magic number 32 found (*low*)
- Line 328: Magic number 33 found (*low*)
- Line 329: Magic number 34 found (*low*)
- Line 330: Magic number 35 found (*low*)
- Line 335: Missing docstring (*medium*)
- Line 353: Magic number 50 found (*low*)

Suggestions & Fixes

Fix Critical Bugs

There are 2 bugs that need immediate attention

Security Vulnerabilities

Found 5 security issues

Performance Optimizations

Consider fixing 1 performance issues

Code Style Improvements

Found 84 style issues to address

Suggested Fixes

Bug Fix

Original: Line 60: Potential infinite loop detected

Fixed: Fix the logic error

Explanation: Potential infinite loop detected

Bug Fix

Original: Line 164: Division by zero detected

Fixed: Fix the logic error

Explanation: Division by zero detected

Security Fix

Original: Line 16: Hardcoded credential found

Fixed: Fix the security vulnerability

Explanation: Hardcoded credential found

Security Fix

Original: Line 52: Potential Command Injection vulnerability

Fixed: Fix the security vulnerability

Explanation: Potential Command Injection vulnerability